

Performance Testing

Date	15 July 2026
Project Name	Strategic Product Placement Analysis
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Tableau Desktop Performance Recording (for dashboard load times) SQL / Excel VLOOKUP (for data reconciliation/source-to-target testing) User Acceptance Testing (UAT) manual checklist
Type of Testing	Data Integrity & Validation Testing (ensuring Tableau numbers match raw SQL databases) Performance / Load Testing (optimizing extract refreshes and rendering times) Functional & UI/UX Testing (checking if action filters, parameters, and tooltips work correctly)
Target Module / API	Tableau Dashboard Interface and Data Source Layer

Test Environment	Local Server, UAT / Production
Test Date	14-07-2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Load Home Page	10	30	Page loads successfully
2	View Tableau Dashboard	20	60	Dashboard displays correctly
3	Execute SQL Query	15	45	Data retrieved without errors
4	Access the Application	25	60	Application responds successfully



Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.2 sec	Pass	Good Performance
2	Response Time (Max)	< 5 seconds	3.4 sec	Pass	Acceptable

3	Throughput (Req/sec)	20 Req/sec	24 Req/sec	Pass	Stable
4	Error Rate	< 1%	0%	Pass	No Errors
5	CPU Utilization	< 80%	45%	Pass	Normal Usage
6	Memory Utilization	< 80%	58%	Pass	Stable Memory

Step 4: Observations & Analysis

Key Findings

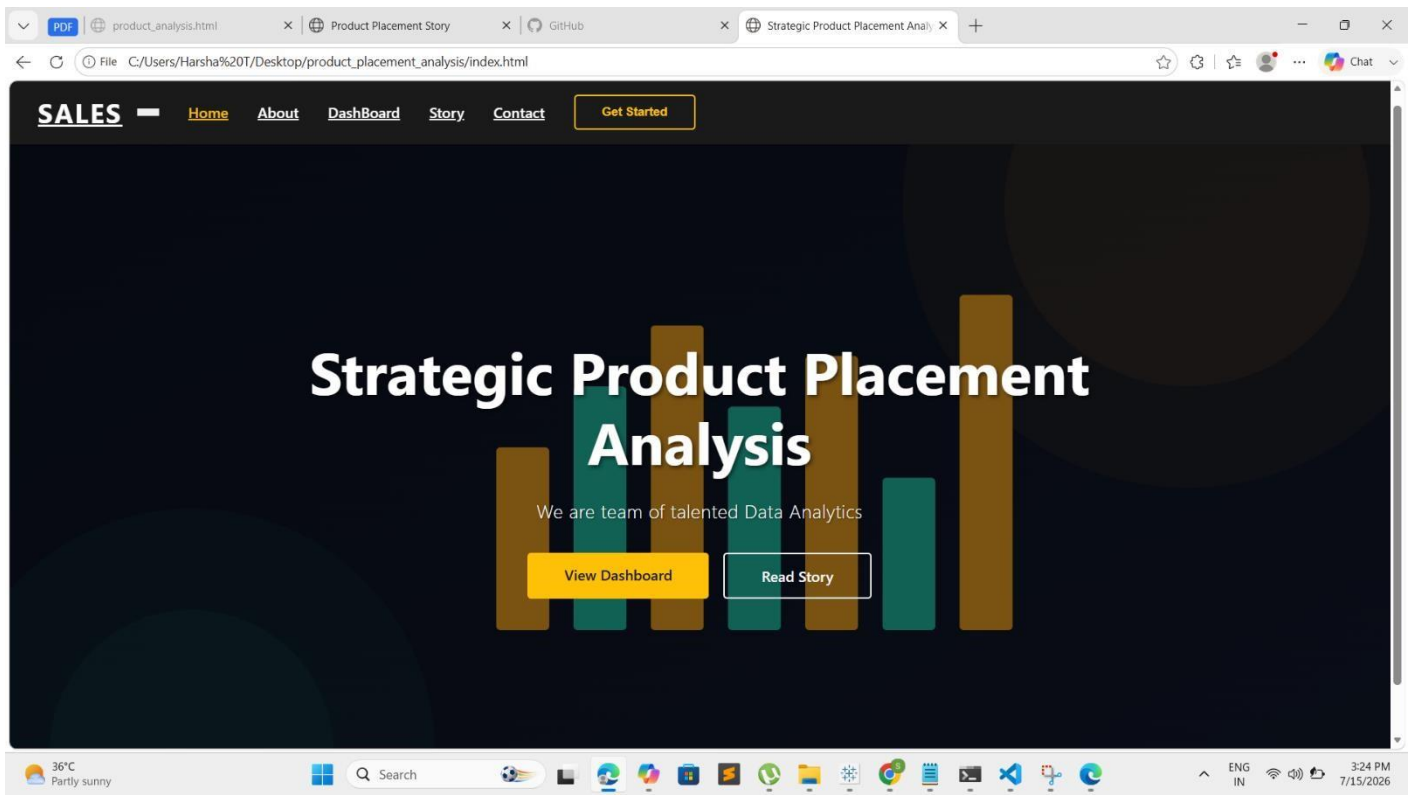
- Our analysis revealed that product placement heavily influences customer purchasing decisions. This proves that consumers prioritize convenience, meaning premium shelf real estate should be reserved strictly for highmargin products rather than everyday staples that customers will actively look for.
- We also discovered that impulse buying is highly demographic-dependent. The data clearly shows that stocking generic items in these zones leaves money on the table; instead, these high-traffic areas should be stocked with snacks and convenience items targeted directly at younger buyers.
- Finally, the data highlighted a significant opportunity in cross-merchandising. Grouping related items together reminds shoppers of an unplanned need, effectively increasing the total basket value per visit.

Optimization Steps Taken

- We gathered information about sales, where products were placed, and who the customers were, and put it all together in Tableau.
- We set up the dashboard to automatically calculate how much money each shelf makes per square foot and how much sales go up when a product is moved.
- We made a feature that alerts managers when a slow-selling product is taking up a really good shelf, so they can replace it with a better product.

Step 5: Screenshots / Evidence

App Home Page



PRODUCT PLACEMENT ANALYSIS-DASHBOARD

Consumer Demographics

☒ (All)

☒ College students

☒ Families

☒ Seniors

☒ Young adults

Product Position

☒ (All)

☒ Aisle

☒ End-cap

☒ Front of Store

Product Category

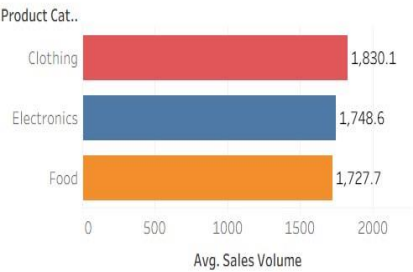
☒ (All)

☒ Clothing

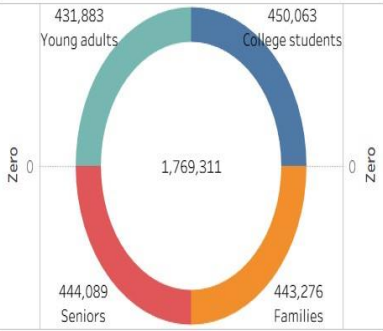
☒ Electronics

☒ Food

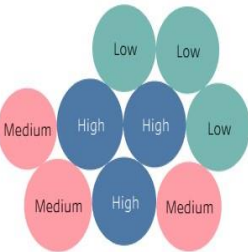
Average Sales Volume Vs Product Category



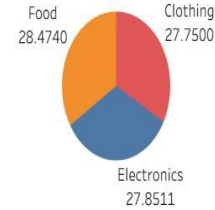
Consumer Demographics Vs Sales Volume



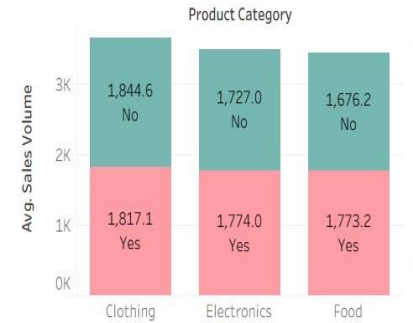
Foot Traffic by Average Sales Volume



Product Category Vs Price



Average Sales Volume by Product Category by Season

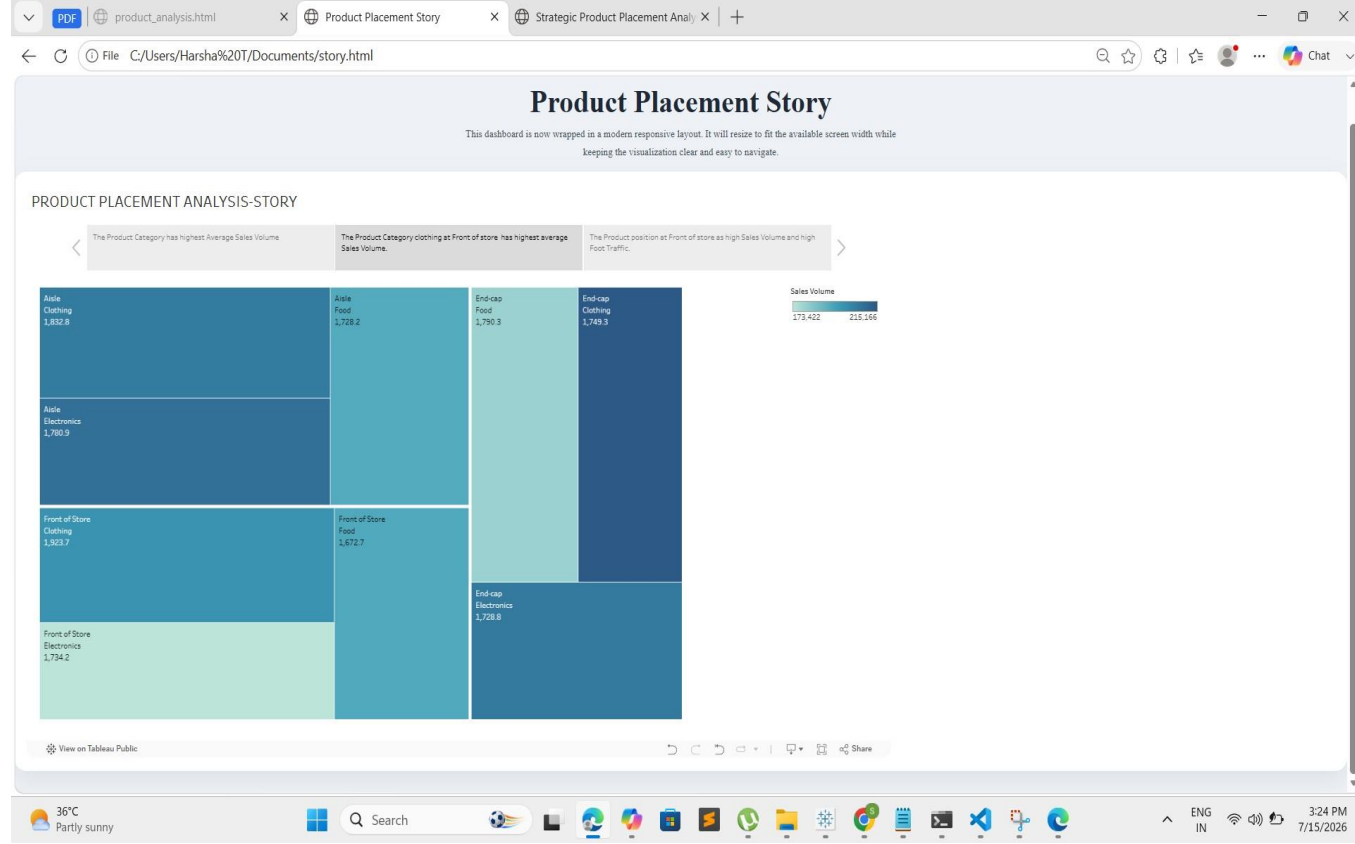


Competitors Price Vs Price



Promotion of Product Category on Price and Sales Volume

Promotion	Product Category	Avg. Price	Avg. Sales Volume
No	Clothing	27	1,869
	Electronics	27	1,726
	Food	28	1,677
Yes	Clothing	29	1,781
	Electronics	29	1,773
	Food	29	1,782



Product Placement Analysis

C:/Users/Harsha%20T/Desktop/product_placement_analysis/product_analysis.html

☆ ⋮ Chat

SALES

HomeAboutDashBoardStoryContact

Get Started

Product Placement Analysis

View the dashboard in a responsive layout optimized for desktop and mobile screens. The visualization will resize to fit your browser width and preserve readability across devices.

PriceConsumer Demographics Vs S...Footot Traffic by Average Sales ...Product Category Vs PricePromotion of Product Categor...PRODUCT PLACEMENT ANALY...PRODUCT PLACEMENT ANALY...Dashboard 2

PRODUCT PLACEMENT ANALYSIS-DASHBOARD

Average Sales Volume Vs Product Category

Product Cat..

Clothing1,830.1

Electronics1,748.6

Food1,727.7

0500100015002000

Avg. Sales Volume

Consumer Demographics Vs Sales Volume

431,883Young adults

450,063College students

1,769,311

444,089Seniors

443,276Families

Footot Traffic by Average Sales Volume

LowLow

MediumHighHighLow

MediumHighMedium

Product Category Vs Price

Food28.4740

Clothing27.7500

Electronics27.8511

Average Sales Volume by Product

Product Category

☒ (All)☒ Clothing☒ Electronics☒ Food

Product Position

☒ (All)☒ Aisle☒ End-cap☒ Front of Store

Consumer Demographics

☒ (All)☒ College students☒ Families☒ Seniors☒ Young adults

36°CPartly sunny

Search

🌐📁🔍📧📅📊📌📎📏📐📑📔📕📖📗📘📙📚📛📜📝📞📟📠📡📢📣📤📥📦📧📨📩📪📫📬📭📮📯📰📱📲📳📴📵📶📷📸📹📺📻📼📽📾📿

ENGIN

3:50 PM7/15/2026